Calendar
REAL VARIABLES I
Fall 2026

I will put all deadlines on your Blackboard Calendar. Please check it regularly!

0.1 Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	First day of classes
Friday, August 21	Last day to add this course through the standard add period
Friday, August 28	Last day to drop without a W
Thursday, December 3	Last day of classes and last day to withdraw from all classes

0.2 Exams

Date	Event
Through October 27	Oral exam 1 appointments must be completed by this date
October 21, 2026	Take-home written midterm due on Gradescope (120 minutes plus 15
Wednesday, December 9, 2026, 10:15 - 12:15	Final-exam period

0.3 Daily lecture log

	Wk	Lec
		L1: Riemann integration and its limitations
2026-08-17	1	Review: the Riemann integral; theorem+proof: continuous functions are Riemann integrabl
		L2: Countability and unbounded functions
2026-08-19		Proposition + proof: continuous functions with L1 norm are not a complete metric space. I
		L3: Countability, the extended reals, and sums of non-negative numbers
2026-08-21		Review: fat rationals; sketch: Riemann integrable functions with respect to L^1 are incomple
		L4: Sums of non-negative numbers
2026-08-24	2	Review: continuity of arbitrary non-negative sums with respect to increasing index sets Cor
		L5: Outer measures generated by covering costs
2026-08-26		Motivation: construct a notion of size from elementary sets whose costs are already known.
		L6: Exercise session: Monotone and dominated convergence for sums
2026-08-28		Exercise session: Monotone and dominated convergence for sums
		L7: Lebesgue outer measure on $\mathbb{R}^d$
2026-08-31	3	Prop+proof: maximality of generated outer measures. Defn: axis-parallel rectangles, volum
		L8: Lebesgue outer measure and the Vitali obstruction
2026-09-02		Prop+proof: translation invariance and dilation scaling of λ^*. Defn: Vitali equivalence relat
		L9: Exercise session: HW02
2026-09-04		Conclusion of proof: rational balls as generators of Borels on $\mathbb{R}^d$ Exercise session: HW02.
		No class: Labor Day
2026-09-07	4	Labor Day holiday; no class.
		L10: Measures and their basic inequalities
2026-09-09	TENTATIVE	Defn: measures and measure spaces. Prop+proof: finite additivity, monotonicity, and the d
		L11: Carathéodory measurable sets, σ-algebras and measures: examples
2026-09-11		Thm+proof(conclusion): Carathéodory measurable sets are a σ algebra and μ^* is a measure
		L12: Dynkin systems and measurable functions
2026-09-14	5	Defns: finite, probability, and σ-finite measures. Defns: π-systems and Dynkin (λ-)systems.
		L13: Measurable functions (continued)
2026-09-16		Continuation of the previous lecture's topic: measurable functions and their operations. On
		L14: Exercise session
2026-09-18		Exercise session.
		L15: The Lebesgue and Archimedean integrals: basic definitions
2026-09-21	6	Defns: superlevel distribution functions, the Archimedean integral, and the non-negative Le
		L16: Monotone convergence and positive linearity
2026-09-23		Prop+proof: linearity of the values-times-measures sum for simple functions. Prop+proof:
		L17: Homework exercise session
2026-09-25		Homework exercise session.
		L18: Properties of the Lebesgue integral: Markov's inequality and null sets
2026-09-28	7	Example: an unbounded function with finite integral, computed by superlevel sets and by n
		L19: Uniform integrability and integrable signed functions
2026-09-30		Defn: uniform integrability over measurable sets. Lemma+proof: uniform integrability of a
		L20: Exercise session: HW06
2026-10-02		Exercise session: HW06 (the non-negative integral, monotone convergence, Fatou's lemma,
		L21: Dominated convergence and Fatou's lemma
2026-10-05	8	Thm (statement): Lebesgue dominated convergence theorem. Prop+proof: dominated conv
		L22: Egorov's theorem and Vitali's convergence theorem
2026-10-07		Thm+proof: Egorov's theorem. Example: failure of Egorov's theorem on an infinite-measur
		L23: Exercise session: HW07
2026-10-09		Exercise session: HW07 (rescaling and the Dirac delta; limits of integrals via dominated cor
		L24: Riemann and Lebesgue integrals: semicontinuity and envelopes
2026-10-12	9	Thm (statement): Lebesgue's criterion for Riemann integrability and agreement of the integ
		L25: Proof of Lebesgue's criterion
2026-10-14		Defn: infimal convolution. Thm+proof: Lipschitz approximation from below of lower semic

TENTATIVE: Future lecture topics and their placement may change as the course develops.